Plate tectonics

18a

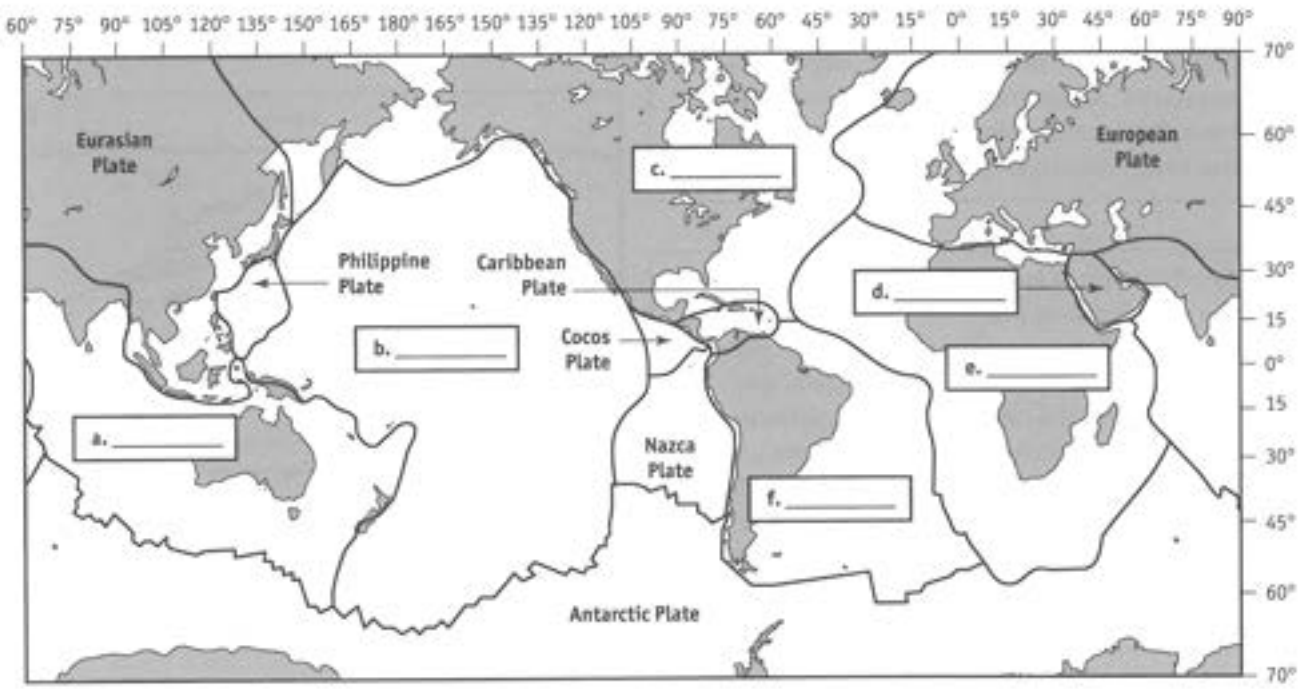
Labelling plates

You may have learnt that the earth is not completely solid and that the lithosphere is not the same thickness all the way through to the mantle. The earth's crust is not even, smooth and continuous but consists of twelve rigid main plates. These tectonic plates move continually over the semi-liquid layer in the mantle, the atmosphere. Geologists have identified the following twelve main tectonic plates:

Australian plate	Antarctic plate	Pacific plate
North American plate	Nazca plate	South American plate
African plate	Eurasian plate	Philippine plate
Caribbean plate	Cocos plate	Arabian plate

1 The map of the world below shows all the major plate boundaries. Only six plates have been labelled.

Label the other six plates. Use the plate names as clues to their locations.



2 Use the text on the next page (question 4) to label the different types of plate boundaries on the map in above. Use coloured pens for the different boundary types and draw a key.

3 Use the information to complete the summary table below.

Type of boundary	Description	Location	Earthquakes, volcanoes or both caused?

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Plate boundaries

All twelve tectonic plates interact with most of their surrounding plates. These interactions take place at the plate boundaries. There are three types of plate boundaries:

- subduction or convergent boundaries
- spreading or divergent boundaries
- transform fault boundaries.

Each type of boundary experiences earthquakes as the plates move past, under or over each other.

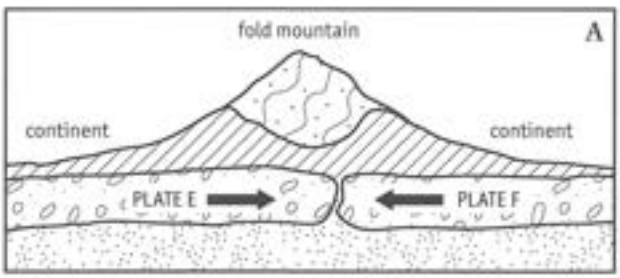
4 Read the information on the different plate boundaries, then match the text to the diagrams by joining them with lines.

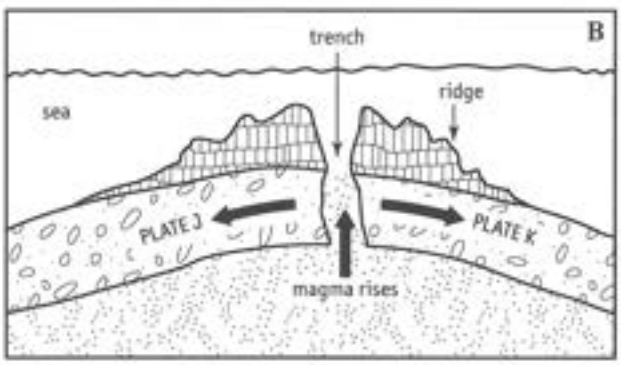
a Subduction or convergent boundaries occur when a thinner oceanic plate collides with a thicker continental plate. The oceanic plate is forced under the continental plate and down into the mantle. The movement of the plate causes great friction. Earthquakes and heat are generated causing the continental crust to melt and form volcanoes at the edge of the continent. An example of this type of plate boundary occurs between the western side of the South American plate and the Nazca plate.

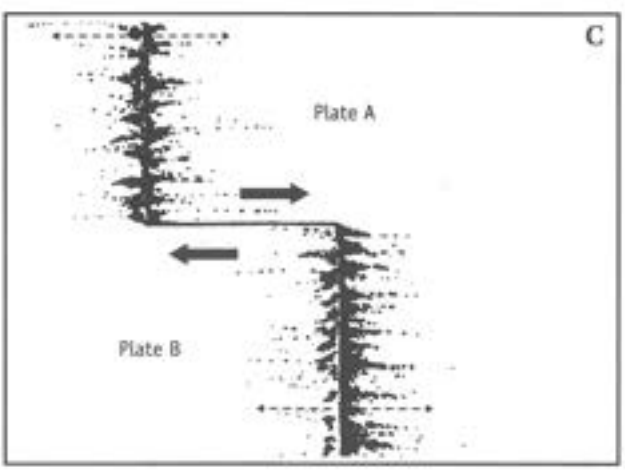
b Another type of subduction boundary is called the collision boundary. This occurs when two continental plates collide. When both plates are of a similar thickness, one plate is not forced under the other. Instead mountains form as the ground is pushed upwards and downwards. An example of this type of plate boundary is the Australian plate colliding with the Eurasian plate forming the Himalayan mountain range.

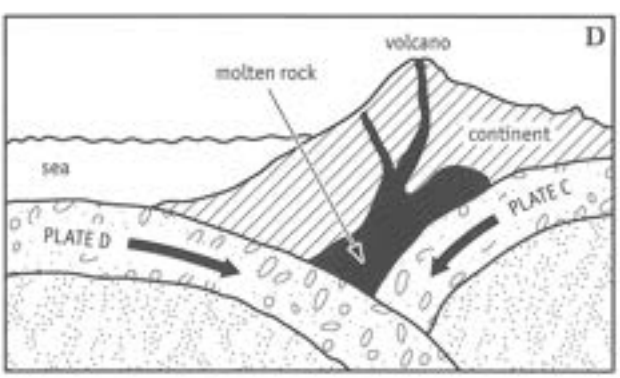
c Divergent boundaries form when two plates move away from each other. The gap that forms between the two plates allows molten material from the mantle to rise and form a chain of volcanoes along the ridges. As the plates move apart, new oceanic crust is produced. All these zones are found under oceans and are called mid-oceanic ridges. An example of a divergent boundary is between the South American plate and the African plate.

d Transform fault boundaries occur when two plates slide past each other forming a transform fault zone. An example of this fault is where the Pacific plate is moving past the North American plate.









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